

Name	Shivang Jhalani
SRN	PES1UG23CS548
SECTION	I

1. Project Structure : MVC Separation

```
Lab4 on / main [!?] is v0.0.1-SNAPSHOT via v21.0.0 via impure (devenv-shell-env)
) tree src
src
├── main
│   ├── java
│   │   ├── com
│   │   │   └── example
│   │   │       ├── studentmanagement
│   │   │       │   ├── controller
│   │   │       │   │   ├── StudentApiController.java
│   │   │       │   │   └── StudentWebController.java
│   │   │       │   ├── model
│   │   │       │   │   └── Student.java
│   │   │       │   ├── repository
│   │   │       │   │   └── StudentRepository.java
│   │   │       │   ├── service
│   │   │       │   │   ├── StudentService.java
│   │   │       │   └── StudentManagementApplication.java
│   └── resources
│       ├── application.properties
│       ├── static
│       ├── templates
│       └── index.html
13 directories, 8 files
```

2. Run the Application

```
Lab4 on / main is v0.0.1-SNAPSHOT via v21.0.0 via impure (devenv-shell-env)
) mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-management-system >-----
[INFO] Building student-management-system 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ student-management-system ---
[INFO] Deleting /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ student-management-system ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ student-management-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 6 source files with javac [debug release 21] to target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ student-management-system ---
[INFO] skip non existing resourceDirectory /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/src/test/resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ student-management-system ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ student-management-system ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ student-management-system ---
[INFO] Building jar: /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/target/student-management-system-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:3.2.3:repackage (repackage) @ student-management-system ---
[INFO] Replacing main artifact /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/target/student-management-system-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/target/student-management-system-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] --- install:3.1.1:install (default-install) @ student-management-system ---
[INFO] Installing /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/pom.xml to /home/shivang/.m2/repository/com/example/student-management-system/0.0.1-SNAPSHOT/student-management-system-0.0.1-SNAPSHOT.pom
[INFO] Installing /home/shivang/shivang/obsidian/notes/sem6/ooad/Lab4/target/student-management-system-0.0.1-SNAPSHOT.jar to /home/shivang/.m2/repository/com/example/student-management-system/0.0.1-SNAPSHOT/student-management-system-0.0.1-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.906 s
[INFO] Finished at: 2026-03-11T17:32:42+05:30
[INFO] -----
```

```
[INFO] skip non existing resourceDirectory /home/shivang/shivang/obsidian/notes/sem6/ooad/lab4/src/test/resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ student-management-system ---
[INFO] No sources to compile
[INFO]
[INFO] <<< spring-boot:3.2.3:run (default-cli) < test-compile @ student-management-system <<<
[INFO]
[INFO] --- spring-boot:3.2.3:run (default-cli) @ student-management-system ---
[INFO] Attaching agents: []

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=====|_|=====
:: Spring Boot ::                (v3.2.3)

2026-03-11T17:36:25.807+05:30 INFO 188055 --- [Student Management System] [main] c.e.s.StudentManagementApplication : Starting StudentManagementApplication using Java 21.0.8 with PID 1080
SS (/home/shivang/shivang/obsidian/notes/sem6/ooad/lab4/target/classes started by shivang in /home/shivang/shivang/obsidian/notes/sem6/ooad/lab4)
2026-03-11T17:36:25.809+05:30 INFO 188055 --- [Student Management System] [main] c.e.s.StudentManagementApplication : No active profile set, falling back to 1 default profile: "default"
2026-03-11T17:36:26.351+05:30 INFO 188055 --- [Student Management System] [main] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-03-11T17:36:26.411+05:30 INFO 188055 --- [Student Management System] [main] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 40 ms. Found 1 JPA repository interface.
2026-03-11T17:36:26.847+05:30 INFO 188055 --- [Student Management System] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-03-11T17:36:26.858+05:30 INFO 188055 --- [Student Management System] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-03-11T17:36:26.858+05:30 INFO 188055 --- [Student Management System] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.19]
2026-03-11T17:36:26.893+05:30 INFO 188055 --- [Student Management System] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-03-11T17:36:26.894+05:30 INFO 188055 --- [Student Management System] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1035 ms
2026-03-11T17:36:26.932+05:30 INFO 188055 --- [Student Management System] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-03-11T17:36:27.124+05:30 INFO 188055 --- [Student Management System] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:file:./data/studentdb user=PE51UG23C5548
2026-03-11T17:36:27.125+05:30 INFO 188055 --- [Student Management System] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-03-11T17:36:27.134+05:30 INFO 188055 --- [Student Management System] [main] o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available at 'jdbc:h2:file:./data/studentdb'
2026-03-11T17:36:27.258+05:30 INFO 188055 --- [Student Management System] [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-03-11T17:36:27.302+05:30 INFO 188055 --- [Student Management System] [main] org.hibernate.Version : HHH000412: Hibernate ORM core version 6.4.4.Final
2026-03-11T17:36:27.333+05:30 INFO 188055 --- [Student Management System] [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-03-11T17:36:27.487+05:30 INFO 188055 --- [Student Management System] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-03-11T17:36:27.519+05:30 WARN 188055 --- [Student Management System] [main] org.hibernate.orm.deprecation : HHH9000025: H2Dialect does not need to be specified explicitly using 'hibernate.dialect' (remove the property setting and it will be selected by default)
2026-03-11T17:36:28.076+05:30 INFO 188055 --- [Student Management System] [main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
Hibernate: create table students (id bigint generated by default as identity, course varchar(255) not null, email varchar(255) not null, name varchar(255) not null, primary key (id))
Hibernate: alter table if exists students drop constraint if exists UK_e2ndfrsx22acpq2tylcaeyuw
Hibernate: alter table if exists students add constraint UK_e2ndfrsx22acpq2tylcaeyuw unique (email)
2026-03-11T17:36:28.129+05:30 INFO 188055 --- [Student Management System] [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-03-11T17:36:28.129+05:30 INFO 188055 --- [Student Management System] [main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-03-11T17:36:28.481+05:30 INFO 188055 --- [Student Management System] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page template: index
2026-03-11T17:36:28.762+05:30 INFO 188055 --- [Student Management System] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path ''
2026-03-11T17:36:28.768+05:30 INFO 188055 --- [Student Management System] [main] c.e.s.StudentManagementApplication : Started StudentManagementApplication in 3.308 seconds (process running for 3.63)
```

3. HomePage

Student Management System

Student Registration Form

Name

Email

Course

Save Student

Student

ID

http://localhost:8080

ChatGPT — chatgpt.com

YouTube

Inbox (96) - shivang2004halani@gmail.com - Gmail

Google Gemini

detectportal.firefox — http://detectportal.firefox.com/canonical.html

Switch to Tab

Switch to Tab

Switch to Tab

4. Add a Student

Student Management System

Student Registration Form

Student List

ID	Name	Email	Course
1	Shivang Jhalani	pes1ug23cs548@pesu.pes.edu	CSE

5. Add More Students

Student Management System

Student Registration Form

Student List

ID	Name	Email	Course
1	Shivang Jhalani	pes1ug23cs548@pesu.pes.edu	CSE
2	Kshitij Kale	horizanblack@gmail.com	CSE
3	Carol	carol@example.com	Electronics

6. Duplicate Email Rejection

Student Management System

Student Registration Form

Email already registered

Student List

ID	Name	Email	Course
1	Shivang Jhalani	pes1ug23cs548@pesu.pes.edu	CSE
2	Kshitij Kale	horizanblack@gmail.com	CSE
3	Carol	carol@example.com	Electronics

7. REST API

```
mvn spring-boot:run .../sem6/ooad/lab4 x

lab4 on  main [!?] is v0.0.1-SNAPSHOT via v21.0.8 via impure (devenv-shell-env)
> curl -s http://localhost:8080/api/students | python3 -m json.tool
[
  {
    "id": 1,
    "name": "Shivang Jhalani",
    "email": "pes1ug23cs548@pesu.pes.edu",
    "course": "CSE"
  },
  {
    "id": 2,
    "name": "Kshitij Kale",
    "email": "horizanblack@gmail.com",
    "course": "CSE"
  },
  {
    "id": 3,
    "name": "Carol",
    "email": "carol@example.com",
    "course": "Electronics"
  }
]

lab4 on  main [!?] is v0.0.1-SNAPSHOT via v21.0.8 via impure (devenv-shell-env)
> |
```

```

lab4 on  main [!?] is  v0.0.1-SNAPSHOT via  v21.0.8 via  impure (devenv-shell-env)
) curl -s -X POST http://localhost:8080/api/students \
  -H "Content-Type: application/json" \
  -d '{"name":"YourMoM","email":"MOM@example.com","course":"BigData"}'
{"id":4,"name":"YourMoM","email":"MOM@example.com","course":"BigData"}

lab4 on  main [!?] is  v0.0.1-SNAPSHOT via  v21.0.8 via  impure (devenv-shell-env)
)

```

Student Management System

Student Registration Form

Email already registered

Aaloo Gobi

pes1ug23cs548@pesu.pes.edu

CSE

Save Student

Student List

ID	Name	Email	Course
1	Shivang Jhalani	pes1ug23cs548@pesu.pes.edu	CSE
2	Kshitij Kale	horizanblack@gmail.com	CSE
3	Carol	carol@example.com	Electronics
4	YourMoM	MOM@example.com	BigData

8. H2 Console

Auto commit

Max rows: 1000

Auto complete Off

Auto select On

jdbc:h2:file:./data/studentdb

STUDENTS

INFORMATION_SCHEMA

Users

H2 2.2.224 (2023-09-17)

Run

Run Selected

Auto complete

Clear

SQL statement:

SELECT * FROM STUDENTS;

SELECT * FROM STUDENTS;

ID	COURSE	EMAIL	NAME
1	CSE	pes1ug23cs548@pesu.pes.edu	Shivang Jhalani
2	CSE	horizanblack@gmail.com	Kshitij Kale
3	Electronics	carol@example.com	Carol
4	BigData	MOM@example.com	YourMoM

(4 rows, 2 ms)

Edit